SUDIPTA BANIK

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SUMMARY

Software Engineer with 7 years of experience at Uber, Grab, and FourKites specializing in scalable, fault-tolerant distributed systems and event-driven backends. Proficient in Go, Java, SQL, Redis, and Kafka. Drove backend projects end to end, from design to production, across systems handling millions of transactions or events daily, spanning fintech, distributed compute, and global logistics domains. Open to relocation.

WORK EXPERIENCE

UBER

Software Engineer II Oct 2024 - Present
Mobility & Batch Compute Teams Bangalore, India

- **Designed Uber Moto rider preferences** for the core trip path and aligned service owners across teams on the approach before implementing the rider and driver backend changes. Rolled it out city by city across LATAM through experiments.
- **Led migration of a Kubernetes job-status service from zonal to regional Cassandra**, removing a per-cluster provisioning burden on the Cassandra team. Validated the data by having the background resource-usage service, which feeds financial reporting, compare reads from both databases and emit mismatch metrics.
- **Made the YARN router tolerate a failed cluster**: added per-cluster call timeouts and a ZooKeeper-backed soft-block, so a single degraded cluster no longer failed requests for clients like Spark, which now received partial results from the healthy clusters.
- **Designed and built Cadence-based test-gated deployments for Apache YARN**, where a build that passes validation on a low-tier cluster triggers the existing pipeline without manual approval, saving manual release effort across the team.
- **Built a policy-based admission layer in the batch Kubernetes federator** that checks workloads at creation against configurable rules served from a cache, then helped extend it for a separate project that reused it.

GRAB

Senior Software Engineer Oct 2023 - Oct 2024
Lending Core Team, Fintech Bangalore, India

- Redesigned the loan offer and credit limit generation from a monthly batch job to **real-time event-driven processing using Kafka**, reducing peak DB load by ~30% and reducing loan offer waiting time for **millions of drivers**.
- Reduced loan creation API p95 latency by almost 32% via **MySQL batch inserts** for instalment. The reduction in network round-trips and lock acquisitions **improved throughput during high-concurrency database writes**.
- Developed a **sub-second data aggregator API** integrating 5+ internal APIs for banking partners; with **partial response fallback for high availability**.
- **Improved the internal Kafka consumer framework reliability** via graceful shutdown re-queuing, reducing message loss to near zero across financial transaction pipelines.
- **Mentored 2 new engineers**, guiding them through internal systems, code reviews, and initial deliverables to maintain team velocity and quality.

Software Engineer Oct 2021 - Sep 2023
Lending PayLater Team, Fintech Bangalore, India

- For Grab's Indonesia PayLater launch, **designed and scaled** the **Refund API** to handle 10% of the country's user base. Used **idempotent and state machine design patterns**, automating retries, error recovery and complex refund flows, which handled 3M+ users gracefully.
- Built a **tiered service fee module for real-time charge APIs and batch billing systems**, implementing slab-rate logic per Indonesian regulatory requirements.
- Designed a configurable credit risk assessment module, **collaborating with product teams and external credit bureaus** to integrate country-specific requirements across **multiple Southeast Asian markets**.

- Developed a **robust lending credit score API**, integrating credit score from data-science team models and user metadata services, ensuring **adherence to REST API design and security best practices**.
- Improved CI/CD build speeds by ~18% and increased unit test coverage by ~35% through Go version upgrade, glide to go mod adoption, and systematic refactoring following **clean code adherence and dependency inversions**.

FOURKITES INC.

Software Engineer Jun 2019 - Oct 2021
Multimodal Supply Chain Visibility Team Chennai, India

- Architected a **Redis-backed caching layer to enable direct integration** with 10+ global maritime carriers, reducing reliance on costly third-party data providers. Designed carrier-specific configs to comply with **API rate limits and built end-to-end data parsers and update flows**.
- Improved **global address search API availability** by replacing a shared Elasticsearch dependency with a dedicated Port Autocomplete API utilising **composite SQL indexing and in-memory cache** for 80,000+ ports and achieved **sub-100ms latency**.
- **Designed an async Kafka-based maritime event enrichment pipeline** for ETA/ETD updates by **integrating data from multiple internal microservices** into a unified callback payload ensuring event persistence, configurability, and reliable real-time shipment update delivery to customers.
- Eliminated error-prone QA processes by replacing SSH-based Ruby script execution with an **internal shipment event simulation tool** with dropdown-driven UI, event-specific data fields, and background worker integration.
- Added ocean-specific features to the shipment cloning module, enabling **domain-specific data replication that improved accuracy and impact** of new client sales demos.

EDUCATION

Jadavpur University
Aug 2015 -- Aug 2019
Bachelor of Engineering *in Electrical Engineering* Kolkata, India
Result: CGPA: 7.91/10

TECHNICAL SKILLS

- **Languages:** Golang, Java, C++, TypeScript, SQL
- **Backend & API:** REST, gRPC, Microservices, Event-Driven Architecture, Domain Driven Design, Spring Boot
- **Messaging & Infrastructure:** Apache Kafka, Kubernetes, Docker, Apache YARN, Linux, Terraform
- **Databases & Caching:** MySQL, PostgreSQL, Cassandra, Redis, MongoDB
- **Cloud & Observability:** AWS, GCP, Datadog, Git

ACHIEVEMENTS

- **The Grab Way Award (Q1 2023)** — Recognized for leading engineering improvements in the Buy Now Pay Later (BNPL) project, enhancing system reliability and supporting business growth.

PROJECTS

Travelling Thief Problem *2018*
Explored metaheuristic approaches to approximate solutions for the Travelling Thief Problem, a composite NP-hard problem coupling TSP and 0/1 knapsack selection, under the guidance of a professor at Indian Statistical Institute, Kolkata. *Source: github.com/baniksudipta/ttp_project*

LANGUAGES

English: Full Professional Proficiency **Bengali:** Native